

Chapter 48

Pulse Count Controller (PCNT)

The pulse count controller (PCNT) is designed to count input pulses.

48.1 Introduction

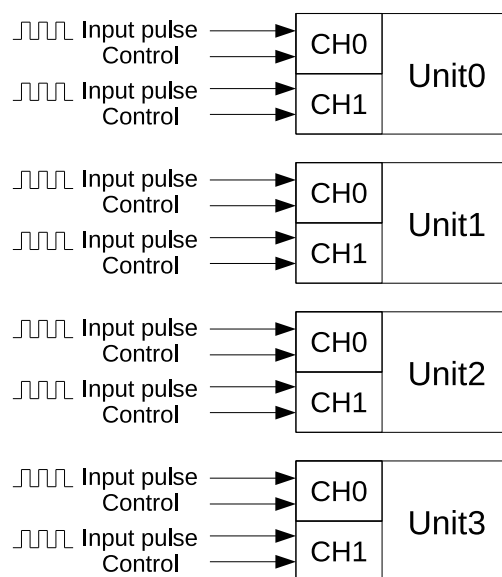


Figure 48.1-1. PCNT Overview

As shown in Figure 48.1-1, PCNT has four independent pulse counters called units, which have their groups of registers. Each unit includes two channels (ch0 and ch1) and a 16-bit signed counter. Each channel has an input pulse signal and an input signal filtering module. The 16-bit signed counter can be configured for incremental or decremental counting. The clock for PCNT module is APB_CLK.

48.2 Feature List

A PCNT has the following features:

- Four independent pulse counters (units) that count from 1 to 65535
- Each unit consists of two independent channels sharing one pulse counter
- All channels have input pulse signals (e.g., sig_ch0_un) with their corresponding control signals (e.g., ctrl_ch0_un)
- Independently filter glitches of input pulse signals (sig_ch0_un and sig_ch1_un) and control signals (ctrl_ch0_un and ctrl_ch1_un) on each unit

- Each channel has the following parameters:
 - Selection between counting on rising or falling edges of the input pulse signal
 - Configuration to Increment, Decrement, or Disable counter mode for control signal's high and low states
- Maximum frequency of pulses: $\frac{f_{APB_CLK}}{2}$

48.3 Architectural Overview

Each unit of PCNT includes two channels (ch0 and ch1) and the functionality of the two channels is identical. The remainder of the chapter will take channel 0 (ch0) as example and n denotes the number of a unit from 0 ~ 3. Each channel of PCNT has two input signals:

- One input pulse signal (e.g., sig_ch0_un, the input pulse signal for ch0 of unit n ch0)
- One control signal (e.g., ctrl_ch0_un, the control signal for ch0 of unit n ch0)

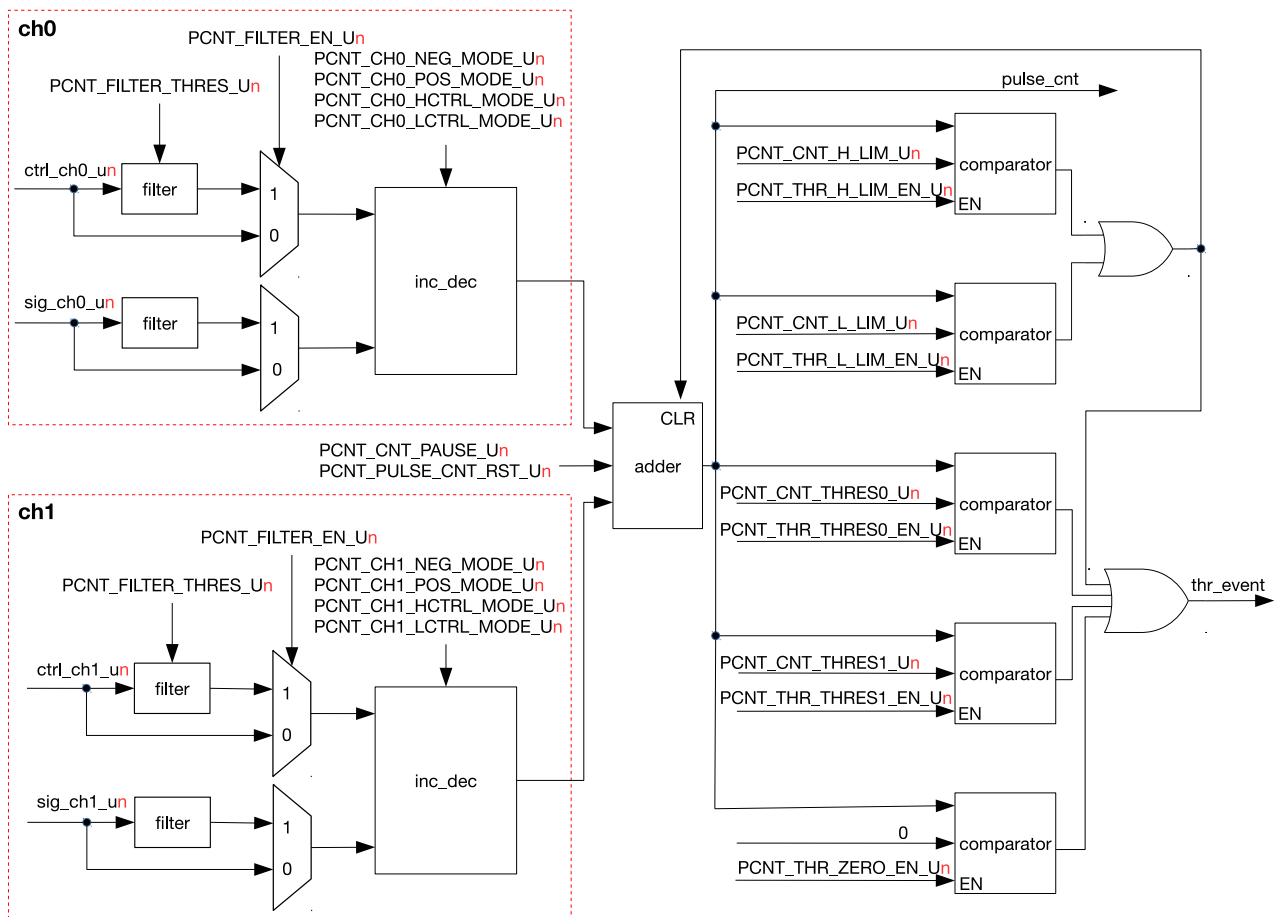


Figure 48.3-1. PCNT Architectural Overview

Figure 48.3-1 shows PCNT's architecture. As stated above, ctrl_ch0_un is the control signal for ch0 of unit n . Its high and low states can be assigned in different counter modes and used for pulse counting of the channel's input pulse signal sig_ch0_un on falling or rising edges.

48.4 Functional Description

The available counter modes of PCNT are as follows:

- Increment mode: When a channel detects an active edge of sig_ch0_un (can be configured by software), the counter value pulse_cnt increases by 1. Upon reaching PCNT_CNT_H_LIM_Un, pulse_cnt is cleared. If PCNT_CNT_PAUSE_Un is set to 1, then pulse_cnt freezes.
- Decrement mode: When a channel detects an active edge of sig_ch0_un (can be configured by software), the counter value pulse_cnt decreases by 1. Upon reaching PCNT_CNT_L_LIM_Un, pulse_cnt is cleared. If PCNT_CNT_PAUSE_Un is set to 1, then pulse_cnt freezes.
- Disable mode: Counting is disabled, and the counter value pulse_cnt freezes.

Table 48.4-1 to Table 48.4-4 provide information on how to configure the counter mode for channel 0.

Table 48.4-1. Counter Mode. Rising edge of Input Pulse Signal. Control Signal in Low State

PCNT_CHO_POS_MODE_Un	PCNT_CHO_LCTRL_MODE_Un	Counter Mode
1	0	Increment
	1	Decrement
	Others	Disable
2	0	Decrement
	1	Increment
	Others	Disable
Others	N/A	Disable

Table 48.4-2. Counter Mode. Rising edge of Input Pulse Signal. Control Signal in High State

PCNT_CHO_POS_MODE_Un	PCNT_CHO_HCTRL_MODE_Un	Counter Mode
1	0	Increment
	1	Decrement
	Others	Disable
2	0	Decrement
	1	Increment
	Others	Disable
Others	N/A	Disable

Table 48.4-3. Counter Mode. Falling Edge of Input Pulse Signal. Control Signal in Low State

PCNT_CHO_NEG_MODE_Un	PCNT_CHO_LCTRL_MODE_Un	Counter Mode
1	0	Increment
	1	Decrement
	Others	Disable
2	0	Decrement
	1	Increment
	Others	Disable
Others	N/A	Disable

Table 48.4-4. Counter Mode. Falling Edge of Input Pulse Signal. Control Signal in High State

PCNT_CHO_NEG_MODE_Un	PCNT_CHO_HCTRL_MODE_Un	Counter Mode
1	0	Increment
	1	Decrement
	Others	Disable
2	0	Decrement
	1	Increment
	Others	Disable
Others	N/A	Disable

Each unit has one filter for all its control and input pulse signals. A filter can be enabled by setting the bit [PCNT_FILTER_EN_Un](#). The filter monitors the signals and ignores all the noise, i.e., the glitches with pulse widths shorter than [PCNT_FILTER_THRES_Un](#) APB clock cycles in length.

As shown on Figure 48.3-1, each unit has two channels which process different input pulse signals and increase or decrease values via their respective inc_dec modules, then the two channels send these values to the counter module which has a 16-bit wide signed register. This counter can be suspended by setting [PCNT_CNT_PAUSE_Un](#), and cleared by setting [PCNT_PULSE_CNT_RST_Un](#).

The PCNT has five watchpoints that share one interrupt. The interrupt can be enabled or disabled by interrupt enable signals of each individual watchpoint.

- Maximum count value: When pulse_cnt reaches [PCNT_CNT_H_LIM_Un](#), a high limit interrupt is triggered and [PCNT_CNT_THR_H_LIM_LAT_Un](#) is high.
- Minimum count value: When pulse_cnt reaches [PCNT_CNT_L_LIM_Un](#), a low limit interrupt is triggered and [PCNT_CNT_THR_L_LIM_LAT_Un](#) is high.
- Two threshold values: When pulse_cnt equals either [PCNT_CNT_THRES0_Un](#) or [PCNT_CNT_THRES1_Un](#), an interrupt is triggered and either [PCNT_CNT_THR_THRES0_LAT_Un](#) or [PCNT_CNT_THR_THRES1_LAT_Un](#) is high respectively.
- Zero: When pulse_cnt is 0, an interrupt is triggered and [PCNT_CNT_THR_ZERO_LAT_Un](#) is valid.

If [PCNT_CNT_H_LIM_Un](#) and/or [PCNT_CNT_L_LIM_Un](#) are reconfigured by software when PCNT is working, the new configuration will take effect after pulse_cnt counts to any of the above five watchpoints; If [PCNT_CNT_THRES0_Un](#) and/or [PCNT_CNT_THRES1_Un](#) are reconfigured by software, the new configuration will take effect immediately.

48.5 Interrupts

ESP32-P4's PCNT can generate the following interrupt signal that will be sent to the [Interrupt Matrix](#).

- [PCNT_INT](#)

There are several internal interrupt sources from PCNT that can generate the above interrupt signal. The interrupt sources from PCNT are listed with their trigger conditions and the resulted interrupt signal in Table 48.5-1.

Table 48.5-1. **PCNT's** Internal Interrupt Sources

Internal Interrupt Source	Trigger Condition	Interrupt Signal
PCNT_CNT_THR_EVENT_U0_INT	reaching the watchpoint in PCNT unit 0	PCNT_INT
PCNT_CNT_THR_EVENT_U1_INT	reaching the watchpoint in PCNT unit 1	PCNT_INT
PCNT_CNT_THR_EVENT_U2_INT	reaching the watchpoint in PCNT unit 2	PCNT_INT
PCNT_CNT_THR_EVENT_U3_INT	reaching the watchpoint in PCNT unit 3	PCNT_INT

Note:

For definitions of *interrupt*, *interrupt signal*, *interrupt source*, and their correlations, please refer to Chapter [12 Interrupt Matrix](#) > Section [12.2 Interrupt Terminology in ESP32-P4](#).

48.6 Programming Procedures

In each unit, channel 0 and channel 1 can be configured to work independently or together. The three subsections below provide details of channel 0 incrementing independently, channel 0 decrementing independently, and channel 0 and channel 1 incrementing together. For other working modes not elaborated in this section (e.g., channel 1 incrementing/decrementing independently, or one channel incrementing while the other decrementing), reference can be made to these three subsections.

48.6.1 Channel 0 Incrementing Independently

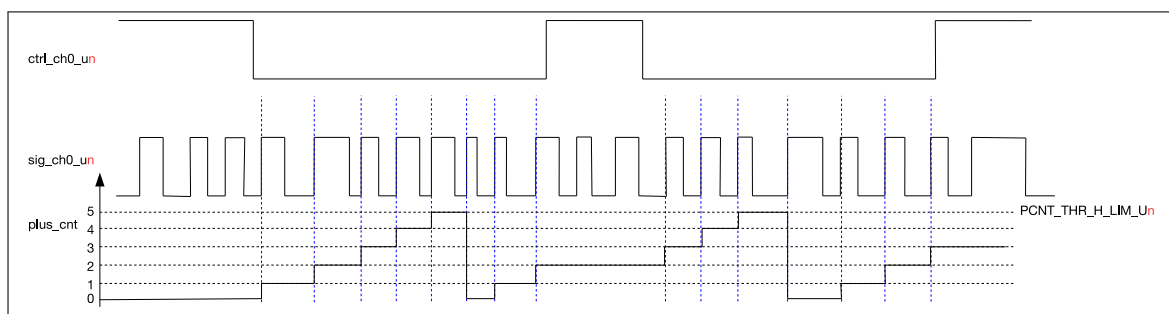


Figure 48.6-1. Channel 0 Up Counting Diagram

Figure [48.6-1](#) illustrates how channel 0 is configured to increment independently on the rising edge of sig_ch0_un while channel 1 is disabled (see Subsection [48.4](#) for how to disable channel 1). The configuration of channel 0 is shown below.

- **PCNT_CHO_LCTRL_MODE_Un=0**: When ctrl_ch0_un is low, the counter mode specified for the low state turns on, in this case it is Increment mode.
- **PCNT_CHO_HCTRL_MODE_Un=2**: When ctrl_ch0_un is high, the counter mode specified for the low state turns on, in this case it is Disable mode.
- **PCNT_CHO_POS_MODE_Un=1**: The counter increments on the rising edge of sig_ch0_un.

- `PCNT_CHO_NEG_MODE_Un=0`: The counter idles on the falling edge of `sig_ch0_un`.
- `PCNT_CNT_H_LIM_Un=5`: When `pulse_cnt` counts up to `PCNT_CNT_H_LIM_Un`, it is cleared.

48.6.2 Channel 0 Decrementing Independently

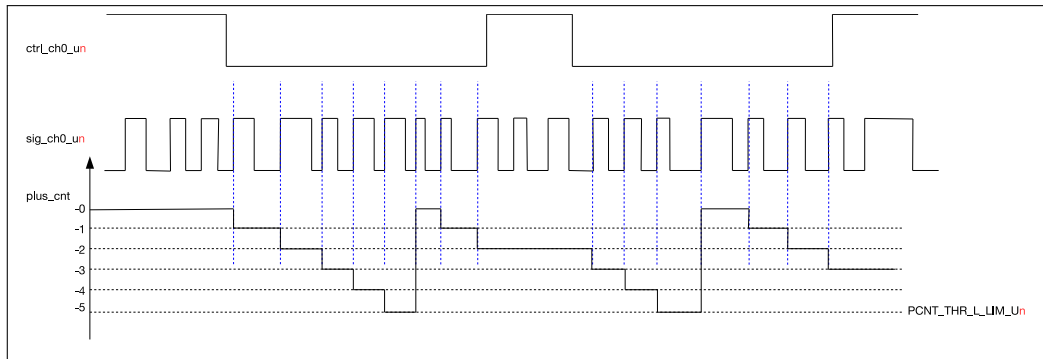


Figure 48.6-2. Channel 0 Down Counting Diagram

Figure 48.6-2 illustrates how channel 0 is configured to decrement independently on the rising edge of `sig_ch0_un` while channel 1 is disabled. The configuration of channel 0 in this case differs from that in Figure 48.6-1 in the following aspects:

- `PCNT_CHO_POS_MODE_Un=2`: the counter decrements on the rising edge of `sig_ch0_un`.
- `PCNT_CNT_L_LIM_Un=-5`: when `pulse_cnt` counts down to `PCNT_CNT_L_LIM_Un`, it is cleared.

48.6.3 Channel 0 and Channel 1 Incrementing Together

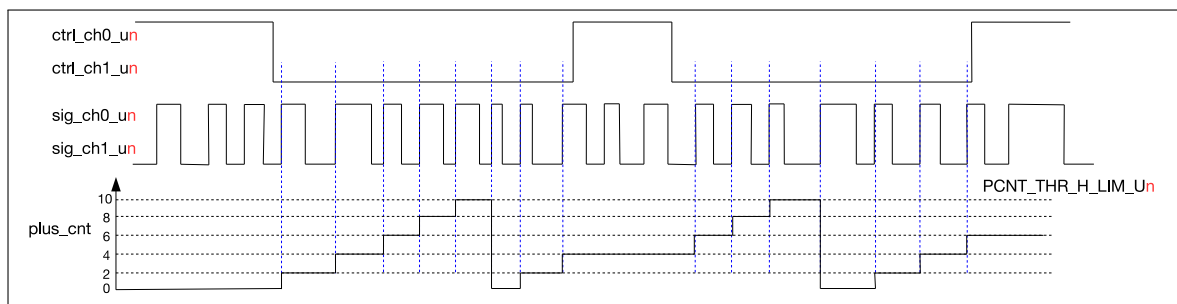


Figure 48.6-3. Two Channels Up Counting Diagram

Figure 48.6-3 illustrates how channel 0 and channel 1 are configured to increment on the rising edge of `sig_ch0_un` and `sig_ch1_un` respectively at the same time. It can be seen in Figure 48.6-3 that control signal `ctrl_ch0_un` and `ctrl_ch1_un` have the same waveform, so as input pulse signal `sig_ch0_un` and `sig_ch1_un`. The configuration procedure is shown below.

- For channel 0:
 - `PCNT_CHO_LCTRL_MODE_Un=0`: When `ctrl_ch0_un` is low, the counter mode specified for the low state turns on, in this case it is Increment mode.

- `PCNT_CHO_HCTRL_MODE_Un`=2: When `ctrl_ch0_un` is high, the counter mode specified for the low state turns on, in this case it is Disable mode.
- `PCNT_CHO_POS_MODE_Un`=1: The counter increments on the rising edge of `sig_ch0_un`.
- `PCNT_CHO_NEG_MODE_Un`=0: The counter idles on the falling edge of `sig_ch0_un`.
- For channel 1:
 - `PCNT_CH1_LCTRL_MODE_Un`=0: When `ctrl_ch1_un` is low, the counter mode specified for the low state turns on, in this case it is Increment mode.
 - `PCNT_CH1_HCTRL_MODE_Un`=2: When `ctrl_ch1_un` is high, the counter mode specified for the low state turns on, in this case it is Disable mode.
 - `PCNT_CH1_POS_MODE_Un`=1: The counter increments on the rising edge of `sig_ch1_un`.
 - `PCNT_CH1_NEG_MODE_Un`=0: The counter idles on the falling edge of `sig_ch1_un`.
- `PCNT_CNT_H_LIM_Un`=10: When `pulse_cnt` counts up to `PCNT_CNT_H_LIM_Un`, it is cleared.

48.7 Register Summary

The addresses in this section are relative to **Pulse Count Controller** base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

Name	Description	Address	Access
Configuration Register			
PCNT_U0_CONFO_REG	Configuration register 0 for unit 0	0x0000	R/W
PCNT_U0_CONF1_REG	Configuration register 1 for unit 0	0x0004	R/W
PCNT_U0_CONF2_REG	Configuration register 2 for unit 0	0x0008	R/W
PCNT_U1_CONFO_REG	Configuration register 0 for unit 1	0x000C	R/W
PCNT_U1_CONF1_REG	Configuration register 1 for unit 1	0x0010	R/W
PCNT_U1_CONF2_REG	Configuration register 2 for unit 1	0x0014	R/W
PCNT_U2_CONFO_REG	Configuration register 0 for unit 2	0x0018	R/W
PCNT_U2_CONF1_REG	Configuration register 1 for unit 2	0x001C	R/W
PCNT_U2_CONF2_REG	Configuration register 2 for unit 2	0x0020	R/W
PCNT_U3_CONFO_REG	Configuration register 0 for unit 3	0x0024	R/W
PCNT_U3_CONF1_REG	Configuration register 1 for unit 3	0x0028	R/W
PCNT_U3_CONF2_REG	Configuration register 2 for unit 3	0x002C	R/W
PCNT_CTRL_REG	Control register for all counters	0x0060	R/W
Status Register			
PCNT_U0_CNT_REG	Counter value for unit 0	0x0030	RO
PCNT_U1_CNT_REG	Counter value for unit 1	0x0034	RO
PCNT_U2_CNT_REG	Counter value for unit 2	0x0038	RO
PCNT_U3_CNT_REG	Counter value for unit 3	0x003C	RO
PCNT_U0_STATUS_REG	PNCT UNIT0 status register	0x0050	RO
PCNT_U1_STATUS_REG	PNCT UNIT1 status register	0x0054	RO
PCNT_U2_STATUS_REG	PNCT UNIT2 status register	0x0058	RO
PCNT_U3_STATUS_REG	PNCT UNIT3 status register	0x005C	RO
Interrupt Register			
PCNT_INT_RAW_REG	Interrupt raw status register	0x0040	RO
PCNT_INT_ST_REG	Interrupt status register	0x0044	RO
PCNT_INT_ENA_REG	Interrupt enable register	0x0048	R/W
PCNT_INT_CLR_REG	Interrupt clear register	0x004C	WO
Version Register			
PCNT_DATE_REG	PCNT version control register	0x00FC	R/W

48.8 Registers

The addresses in this section are relative to **Pulse Count Controller** base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program reserved fields, please refer to Section [Programming Reserved Register Field](#).

Register 48.1. PCNT Un CONFO REG (n: 0-3) (0x0000+0xC*n)

PCNT_CH1_LCTRL_MODE_UO																												PCNT_CH1_HCTRL_MODE_UO																												PCNT_CH1_POS_MODE_UO																												PCNT_CH1_NEG_MODE_UO																												PCNT_CH0_LCTRL_MODE_UO																												PCNT_CH0_HCTRL_MODE_UO																												PCNT_CH0_POS_MODE_UO																												PCNT_CH0_NEG_MODE_UO																												PCNT_THR_THRES1_EN_UO																												PCNT_THR_THRES0_EN_UO																												PCNT_THR_L_LIM_EN_UO																												PCNT_THR_H_LIM_EN_UO																												PCNT_FILTER_EN_UO																												PCNT_FILTER_THRES_UO																											
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9																	0																																																																																																																																																																																																																																																																																																																																																																
0x0		0x0		0x0		0x0		0x0		0x0		0x0		0x0		0		0		1		1		1		1		0x10																Reset																																																																																																																																																																																																																																																																																																																																																											

PCNT_FILTER_THRES_Un Configures the maximum threshold for the filter. Any pulses with width less than this will be ignored when the filter is enabled.

Measurement unit: APB CLK cycles.

(R/W)

PCNT_FILTER_EN_U_n This is the enable bit for unit *n*'s input filter. (R/W)

PCNT_THR_ZERO_EN_U_n This is the enable bit for unit *n*'s zero comparator. (R/W)

PCNT_THR_H_LIM_EN_*Un* This is the enable bit for unit *n*'s thr_h_lim comparator. Configures it to enable the high limit interrupt. (R/W)

PCNT_THR_L_LIM_EN_*n* This is the enable bit for unit *n*'s thr_l_lim comparator. Configures it to enable the low limit interrupt. (R/W)

PCNT_THR_THRES0_EN_U n This is the enable bit for unit n 's thres0 comparator. (R/W)

PCNT_THR_THRES1_EN U_n This is the enable bit for unit n 's thres1 comparator. (R/W)

PCNT_CHO_NEG_MODE_Un Configures the behavior when the signal input of channel 0 detects a falling edge.

1: Increment the counter

2: Decrement the counter

0. 3: No effect

(R/W)

PCNT_CHO_POS_MODE_Un Configures the behavior when the signal input of channel 0 detects a rising edge.

1: Increment the counter

2: Decrement the counter

0. 3: No effect

(R/W)

PCNT_CHO_HCTRL_MODE_U n Configures how the CH n _POS_MODE/CH n _NEG_MODE settings will be modified when the control signal is high.

0: No modification

1: Invert behavior (increase $\rightarrow$ decrease, decrease $\rightarrow$ increase)

2.3: Inhibit counter modification

(R/W)

Continued on the next page...

Register 48.1. PCNT_UN_CONFO_REG (*n*: 0-3) (0x0000+0xCn*)**

Continued from the previous page...

PCNT_CHO_LCTRL_MODE_UN Configures how the CH*n*_POS_MODE/CH*n*_NEG_MODE settings will be modified when the control signal is low.

0: No modification

1: Invert behavior (increase → decrease, decrease → increase)

2, 3: Inhibit counter modification

(R/W)

PCNT_CH1_NEG_MODE_UN Configures the behavior when the signal input of channel 1 detects a falling edge.

1: Increment the counter

2: Decrement the counter

0, 3: No effect

(R/W)

PCNT_CH1_POS_MODE_UN Configures the behavior when the signal input of channel 1 detects a rising edge.

1: Increment the counter

2: Decrement the counter

0, 3: No effect

(R/W)

PCNT_CH1_HCTRL_MODE_UN Configures how the CH*n*_POS_MODE/CH*n*_NEG_MODE settings will be modified when the control signal is high.

0: No modification

1: Invert behavior (increase → decrease, decrease → increase)

2, 3: Inhibit counter modification

(R/W)

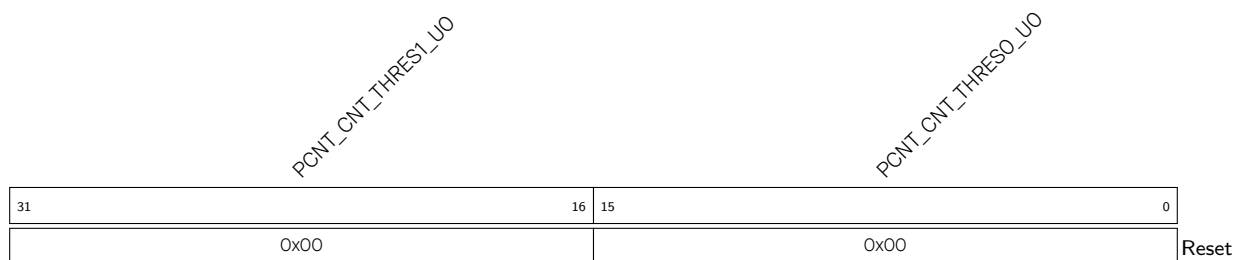
PCNT_CH1_LCTRL_MODE_UN Configures how the CH*n*_POS_MODE/CH*n*_NEG_MODE settings will be modified when the control signal is low.

0: No modification

1: Invert behavior (increase → decrease, decrease → increase)

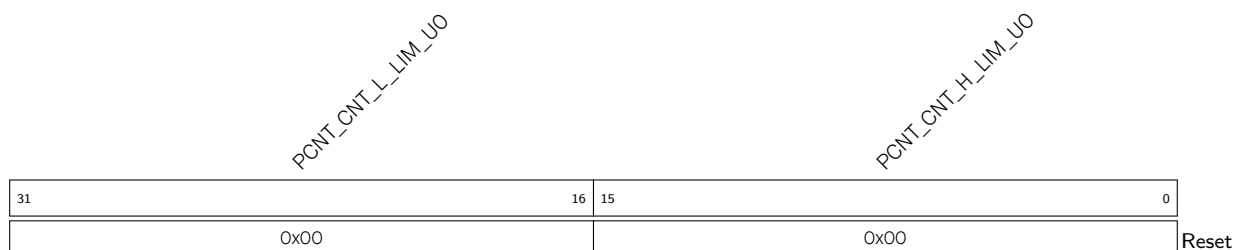
2, 3: Inhibit counter modification

(R/W)

Register 48.2. PCNT_UN_CONF1_REG (*n*: 0-3) (0x0004+0xCn*)**

PCNT_CNT_THRES0_UN Configures the thres0 value for unit *n*. (R/W)

PCNT_CNT_THRES1_UN Configures the thres1 value for unit *n*. (R/W)

Register 48.3. PCNT_UN_CONF2_REG (*n*: 0-3) (0x0008+0xCn*)**

PCNT_CNT_H_LIM_UN Configures the thr_h_lim value for unit *n*. When pulse_cnt reaches this value, the counter will be cleared to 0. (R/W)

PCNT_CNT_L_LIM_UN Configures the thr_l_lim value for unit *n*. When pulse_cnt reaches this value, the counter will be cleared to 0.(R/W)

Register 48.4. PCNT_CTRL_REG (0x0060)

(reserved)																PCNT_CLK_EN		(reserved)		PCNT_CNT_PAUSE_U3 PCNT_PULSE_CNT_RST_U3 PCNT_CNT_PAUSE_U2 PCNT_PULSE_CNT_RST_U2 PCNT_CNT_PAUSE_U1 PCNT_PULSE_CNT_RST_U1 PCNT_CNT_PAUSE_U0 PCNT_PULSE_CNT_RST_U0																
31																17	16	15									8	7	6	5	4	3	2	1	0	Reset
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	1	0	

PCNT_PULSE_CNT_RST_U n Write 1 to clear unit n 's counter.

0: No effect

1: Clear

(R/W)

PCNT_CNT_PAUSE_U n Write 1 to freeze unit n 's counter.

0: No effect

1: Freeze

(R/W)

PCNT_CLK_EN Configures whether or not to enable the registers clock gate of PCNT module.

0: the clock for registers is enabled when registers are read and written

1: the clock for registers is always on

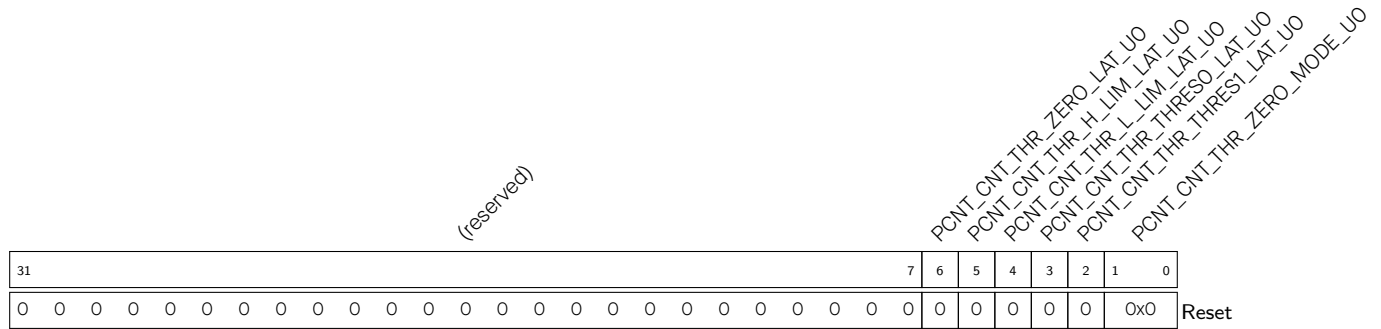
(R/W)

Register 48.5. PCNT_U n _CNT_REG (n : 0-3) (0x0030+0x4* n)

(reserved)																PCNT_PULSE_CNT_U0																		
31																16	15																	0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0x00																Reset		

PCNT_PULSE_CNT_U n Represents the current pulse count value for unit n . (RO)

Register 48.6. PCNT_UN_STATUS_REG (n: 0-3) (0x0050+0x4*n)



PCNT_CNT_THR_ZERO_MODE_UN Represents the pulse counter status of PCNT_UN corresponding to 0.

- 0: pulse counter decreases from positive to 0
 - 1: pulse counter increases from negative to 0
 - 2: pulse counter is negative
 - 3: pulse counter is positive
- (RO)

PCNT_CNT_THR_THRES1_LAT_UN Represents whether the current value of PCNT_UN equals to thres1 when thres1 comparator is enabled.

- 0: others
 - 1: the current pulse counter equals to thres1 and thres1 event is valid
- (RO)

PCNT_CNT_THR_THRES0_LAT_UN Represents whether the current value of PCNT_UN equals to thres0 when thres0 comparator is enabled.

- 0: others
 - 1: the current pulse counter equals to thres0 and thres0 event is valid
- (RO)

PCNT_CNT_THR_L_LIM_LAT_UN Represents whether the current value of PCNT_UN is equal to the low limit threshold when the low limit comparator is enabled.

- 0: others
 - 1: the current pulse counter equals to thr_l_lim and low limit event is valid.
- (RO)

PCNT_CNT_THR_H_LIM_LAT_UN Represents whether the current value of PCNT_UN is equal to the high limit threshold when the high limit comparator is enabled.

- 0: others
 - 1: the current pulse counter equals to thr_h_lim and high limit event is valid.
- (RO)

PCNT_CNT_THR_ZERO_LAT_UN Represents whether the current value of PCNT_UN is 0 when the over-zero comparator is enabled.

- 0: others
 - 1: the current pulse counter equals to 0 and zero threshold event is valid.
- (RO)

Register 48.7. PCNT_INT_RAW_REG (0x0040)

(reserved)																												PCNT_CNT_THR_EVENT_U3_INT_RAW PCNT_CNT_THR_EVENT_U2_INT_RAW PCNT_CNT_THR_EVENT_U1_INT_RAW PCNT_CNT_THR_EVENT_U0_INT_RAW					
31																												4	3	2	1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

PCNT_CNT_THR_EVENT_U n _INT_RAW The raw interrupt status of the PCNT_CNT_THR_EVENT_U n _INT interrupt. (RO)

Register 48.8. PCNT_INT_ST_REG (0x0044)

(reserved)																												PCNT_CNT_THR_EVENT_U3_INT_ST PCNT_CNT_THR_EVENT_U2_INT_ST PCNT_CNT_THR_EVENT_U1_INT_ST PCNT_CNT_THR_EVENT_U0_INT_ST				
31																												4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

PCNT_CNT_THR_EVENT_U n _INT_ST The masked interrupt status of the PCNT_CNT_THR_EVENT_U n _INT interrupt. (RO)

Register 48.9. PCNT_INT_ENA_REG (0x0048)

(reserved)																												PCNT_CNT_THR_EVENT_U3_INT_ENA PCNT_CNT_THR_EVENT_U2_INT_ENA PCNT_CNT_THR_EVENT_U1_INT_ENA PCNT_CNT_THR_EVENT_U0_INT_ENA				
31																												4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

PCNT_CNT_THR_EVENT_U n _INT_ENA Write 1 to enable the PCNT_CNT_THR_EVENT_U n _INT interrupt. (R/W)

